

Family-Centered AI: Building and Sustaining Connections in Multi-Generational Contexts

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The rise of artificial intelligence (AI) has led to a number of questions regarding AI's impact on relationships and the renegotiation of roles, responsibilities, and authority within family settings. Different members of a family may engage with AI systems at different levels, for different purposes, and with different frequencies. Younger family members may have consistent interaction with AI in educational settings, whereas older family members who are retired or working with technology less often may have little to no interactions with AI. Additionally, at a time when younger family members (children and grandchildren) are growing up surrounded by AI technology while older members (parents and grandparents) are not, the nature of family dynamics and opinions around the adoption of AI technology remain underexplored. We define Family-Centered AI as an approach that emphasizes and uplifts existing relationships within the family unit as a whole, while also seeking to address and satisfy the needs of individual members. This work contributes to conceptualizing and defining Family-Centered AI and examining AI's nuanced impacts on family units and individual members across contexts and use cases.

Additional Key Words and Phrases: Intergenerational support, Aging in Place, AI, HCI

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1 Introduction

Family dynamics continually evolve as emerging technologies become embedded in everyday life. AI introduces new complexities into these dynamics because its use is situated within existing relationships, roles, expectations, and patterns of interaction among family members. We conceptualize Family-Centered AI as an approach that moves beyond the individual user to consider how AI use is embedded within, and may shape, the broader relational context of the family. Family members may differ in their needs, expectations, experiences with technology, and preferences for how AI should be used, creating both opportunities for support and potentials for tension within existing relationships. Understanding Family-Centered AI therefore requires examining how families negotiate AI use and how these interactions may shape communication, decision-making, trust, support, and individual agency across family relationships. Designing for family contexts requires attending to these relational processes while recognizing the priorities of both individual family members and the family unit as a whole.

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2 Family-Centered Technology

Before the introduction of the telephone, family members communicated with each other across distances through analog means (i.e., letters) if they did not reside in the same household. As communication technology has evolved from telephones to landlines and eventually cellphones, long-distance, instantaneous communication has become much easier for families [5, 14, 25]. In the present day, AI technology can be used to encourage and ease communication access with families, whether through translation support, companionship technology, or virtual reality tools [24]. However, it remains unclear how families negotiate boundaries and preferences with these technologies across individual members and households.

2.1 The Role of AI Changes in Different Family Contexts

As previously mentioned, AI can be a valuable resource for translation support in multicultural families, whereas monolingual families may envision using AI for support in other areas. Importantly, different families have varying levels of access to technology. Where some households have multiple computers, personal devices, and mobile phones, others may only have a one or two shared devices, or none at all. With this in mind, it is paramount to highlight that there is no generalizable role for AI technology across households and family units; rather, AI adoption strategies must focus on individual feasibility and family needs on a case-by-case basis.

Multicultural families also tend to approach intergenerational relationships and technology use based on their respective cultures, values, and norms. For instance, across several cultural groups in Asia and South America, family support more closely resembles a broader network of family and friends, rather than a single caregiver or location, expanding the size of the family unit [4]. Furthermore, multicultural families often approach conflicts and ambiguous situations between family members differently from monocultural families [21]. Although prior work has examined how to better align LLMs with different cultural values and norms [7, 8], cross-cultural disconnects may still exist, causing user frustration. For example, some family members may become upset with an AI tool if it does not respond in a culturally sensitive way, or disagree with their family members who use AI for different purposes, further straining relationships within the family [21]. As a result, to better maintain a socially-informed perspective on their users, developers of AI technology for older adults must consider variations in cultural nuances, linguistic differences, and attitudes towards communities and individuals when designing their products.

3 Intergenerational Engagement with AI

3.1 Younger Family Member Engagement

As more schools introduce AI technology [17] in academic management [13, 28], grading [10, 18], monitoring [9, 11], and personalized learning [6, 20], it becomes more likely that younger family members are encountering and engaging with AI technology more frequently. Additionally, prior work has documented that youth turn to AI technology such as chatbots for help-seeking queries and general consultation [2, 3, 19]. With these constant interactions and experiences, younger family members have the opportunity to teach other family members about AI technology, shifting authority on information to the younger generation [16]. Such education may not only help other members become more aware of the benefits and risks of using certain technology, but also strengthen the relationships between older and younger family members, highlighting the increasing importance of AI education and digital literacy among multi-generational households.

3.2 Older Family Member Engagement

Many use cases for AI technology among older adults currently center on physical and socio-emotional health support, as older adults tend to have more specific health needs and weaker support systems than their younger family members. Huang et. al documents older adults using AI-based conversational systems to monitor falls, provide reminders to take medication, and assist with other physical needs, which can be shared with family members and caregivers to provide more personalized care [12]. Older adults have also used AI as a social support for loneliness and isolation when separated from their family members. For instance, AI technologies have been used to facilitate long-distance connections between older adults living in long-term care facilities and their family members and caregivers [24], demonstrating a desire to use technology as both a complement to and occasional substitute for human connection. However, this interpretation also poses further questions about the future role AI can or should play in helping older adults stay connected with family and community members across physical distance.

3.3 Joint Decision-Making and Reflection on AI Technology Adoption

Although younger family members and caregivers are generally more familiar and comfortable with using new technologies than older adults, they are not always open to using AI technology to supplement familial support and caregiving, reflecting general hesitation towards the implementation of AI within family environments [26]. We argue that AI technology adoption should not only be negotiated from the perspective of an individual, but also from that of the family unit as a whole, preserving individual member agency. To help families make more informed decisions on AI adoption and improve experiences with technology as a form of connection, we propose that more intergenerational AI literacy and education programs for families should be developed and implemented within care networks. Additionally, as AI becomes more integrated into homes and care facilities, more research must be conducted to quantify the impact of AI on older adults' long-term relationships with their families.

4 AI Impacts and Tensions With Family Relationship Dynamics

4.1 Negative Impacts From AI-Facilitated Scams

In some cases, AI technology has been used to target and extort older adults for money using voice modulation software. For example, “*grandparent scams*”, or scams where an attacker will call and use deepfake AI technology to impersonate a family member to exploit an older adult for money [22], are becoming more frequent and sophisticated. These attackers capitalize on the older adult's position in society and utilize social engineering tactics with information found online to carry out successful scam attempts [22]. Incidents like these can create tension among family members, negatively impacting their relationships with each other. However, prior work has examined how younger family members in intergenerational relationships with older adult family members can be leveraged to teach older adults protective measures to take precautions against these types of attacks [15].

4.2 Misalignment and Miscommunication on AI Technology Adoption

Family members may also use AI in different ways, creating more opportunities for interpersonal conflict when members disagree on the extent of appropriate technology use [1, 27]. Wen et al. examined the possibilities of Generative AI Agents to support family members in caregiving roles with household management, safety education, academic tutoring, and more [23], where they found that communication between family members was paramount in the successful adoption and management of AI technology in different capacities. These misalignments and miscommunications

present a chance to emphasize communication as a priority for exploring mechanisms in which AI can be used to sustain existing family dynamics and further enable discussions around the topic.

5 Conclusion

We conclude with recommendations on defining family-centered AI as contextual and unique to each family's needs and their accessibility to the technology. We perceive the current space of AI technology deployed in family contexts as one that is highly dependent on the preservation and augmentation of intergenerational, familial relationships. As a result, we suggest implementing certain design choices, including stronger digital risk identification tactics, AI responses that better support individuals' agency, and reminders to interact with family members or trained professionals when navigating conflicts with AI technology. We further recommend increasing access to education on digital safety and information protection practices for older adults and their families to better inform their use of AI technologies, particularly in group-learning settings. Ultimately, our recommendations emphasize that AI should supplement, not replace, existing relational dynamics to best support families with multi-generational members. Designing for Family-Centric AI technologies should thus respect the needs and priorities not only of individual family members, but of the family unit as a whole.

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